

Day 1: Tools for TroubleshootingTheory

Task 1 : List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

No.	Real-life scenario	How the multimeter helps
1.	Computer not powering ON	Measures the PSU output voltage to check whether the power supply is providing the correct voltage.
2.	Laptop/PC battery or charging problem	Measures battery voltage and charger output to determine whether the battery or charging adapter is faulty.
3.	Smartphone not charging	Checks voltage, continuity, and resistance around the charging port/circuit to identify a damaged connection or faulty component.

Task 2 : Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

POST Card – Motherboard Hardware Testing

- A POST (Power-On Self-Test) card is a diagnostic tool used to troubleshoot a computer when it fails to boot or does not display anything on the screen. In the video demonstration, the POST card was connected to the motherboard and displayed **hexadecimal POST codes** while the BIOS was performing its hardware checks.
- The displayed codes changed as different hardware components were initialized, such as the **CPU, RAM, chipset, and other motherboard devices**. When the boot process stopped at a particular code, that code provided a clue about the hardware stage where the problem occurred.
- The POST card therefore helped narrow down the faulty hardware without requiring a normal display output. For example, POST codes can indicate problems related to **memory, CPU initialization, or motherboard hardware**, making the card useful for diagnosing systems that fail to start properly.

Overall, the POST card acts as a hardware diagnostic indicator by showing POST codes that help technicians identify the **stage at which the motherboard's boot process fails**.

Task 3 : Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Hardware Diagnostic Tool – HWMonitor

- For this task, I downloaded and installed **HWMonitor**, a free hardware monitoring and diagnostic tool developed by CPUID. The software was run on my Windows laptop to check the current condition of the system hardware.
- HWMonitor displays important hardware health information such as **CPU temperature, voltage, clock speed, GPU temperature, storage information, and battery details** where supported. These readings are useful for identifying problems such as overheating, abnormal voltage levels, or hardware-related issues.
- After running the software, I captured a screenshot of the main dashboard showing the available system health readings. Any personal or sensitive information visible in the screenshot was cropped or blurred before submission.

Software Used: HWMonitor

Purpose: Monitoring and diagnosing computer hardware health

Operating System: Windows 11

❖ Screenshots :

HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
RAVAL				
ASUSTeK COMPUTER INC. FX506HC				
Temperatures				
\TZ.THRM	55.0 °C	55.0 °C	83.0 °C	
Intel PCH	61.0 °C	57.0 °C	66.0 °C	
Fans				
CPU	3400 RPM	3300 RPM	3800 RPM	
Utilization				
Physical Memory Load	56.0 %	56.0 %	59.0 %	
Available Physical Memory	6.9 GB	6.4 GB	6.9 GB	
Virtual Memory Load	63.8 %	61.0 %	64.1 %	
Available Virtual Memory	6.1 GB	6.0 GB	6.5 GB	
Intel Core i5 11400H				
Voltages				
VID (Max)	1.227 V	1.117 V	1.240 V	
Temperatures				
Cores (Max)	49.0 °C	49.0 °C	88.0 °C	
Package	55.0 °C	53.0 °C	89.0 °C	
Powers				
Package	9.61 W	7.30 W	39.98 W	
IA Cores	6.62 W	4.49 W	35.57 W	
GT	0.17 W	0.00 W	1.42 W	
Power Max (PL1)	65.00 W	65.00 W	65.00 W	
Short Power Max (PL2)	83.00 W	83.00 W	83.00 W	
Currents				
IccMax	74.00 A	74.00 A	74.00 A	
Clocks				
Cores (Max)	4482.2 MHz	3994.2 MHz	4655.8 MHz	
CPU BCLK	99.6 MHz	97.2 MHz	103.5 MHz	
LLC/Ring	3286.9 MHz	3209.2 MHz	3828.1 MHz	
Memory	1612.2 MHz	1555.6 MHz	1655.4 MHz	
Utilization				
Processor	5.4 %	3.4 %	48.4 %	
Micron_2400_MTFDKBA512QFM (232041191B16)				
Temperatures				
Assembly	42.0 °C	42.0 °C	50.0 °C	
Sensor 1	42.0 °C	42.0 °C	50.0 °C	
Counters				
Power-On-Hours	4684 hrs	4684 hrs	4684 hrs	
Power Cycle Count	2622	2622	2622	
Health Status	88 %	88 %	88 %	
Utilization				
Used Space (c:)	67.3 %	67.3 %	67.3 %	
Available Capacity (c:)	83.4 GB	83.4 GB	83.4 GB	
Used Space (d:)	53.1 %	53.1 %	53.1 %	
Available Capacity (d:)	45.8 GB	45.8 GB	45.8 GB	

Sensor	Value	Min	Max
Available Capacity (d:)	45.8 GB	45.8 GB	45.8 GB
Used Space (e:)	93.7 %	93.7 %	93.7 %
Available Capacity (e:)	6.2 GB	6.2 GB	6.2 GB
Activity	0.8 %	0.0 %	79.1 %
 Speed			
Read Rate	0.00 MB/s	0.00 MB/s	140.06 MB/s
Write Rate	0.34 MB/s	0.00 MB/s	39.91 MB/s
Read IOPS	0.00 KIOPS	0.00 KIOPS	3.92 KIOPS
Write IOPS	0.03 KIOPS	0.00 KIOPS	2.91 KIOPS
Average Latency	0.00 ms	0.00 ms	15.92 ms
 NVIDIA GeForce RTX 3050 Laptop GPU			
 Voltages			
GPU	0.637 V	0.637 V	0.788 V
Core Power Supply	20.067 V	20.067 V	21.355 V
Frame Buffer Power Supply	20.530 V	20.517 V	20.772 V
8-PIN #0	20.546 V	20.540 V	20.792 V
 Temperatures			
GPU	50.8 °C	50.3 °C	56.9 °C
Hot Spot	57.5 °C	57.0 °C	63.3 °C
 Powers			
GPU	4.56 W	3.91 W	10.98 W
Core Power Supply	0.60 W	0.47 W	4.61 W
Frame Buffer Power Supply	0.86 W	0.22 W	3.32 W
8-PIN #0	1.46 W	0.81 W	7.88 W
Power Limit	60.00 W	60.00 W	60.00 W
 Currents			
GPU	0.07 A	0.04 A	0.38 A
Core Power Supply	0.03 A	0.02 A	0.22 A
Frame Buffer Power Supply	0.04 A	0.01 A	0.16 A
8-PIN #0	0.07 A	0.04 A	0.38 A
 Counters			
PCIe Lanes Errors	0	0	0
PCIe Replay Counter	0	0	0
PCIe Replay Rollover Cou...	0	0	0
PCIe Fatal Errors	0	0	0
PCIe Non Fatal Errors	0	0	0
PCIe Correctable Errors	0	0	0
PCIe LCRC Errors	0	0	0
PCIe Receiver Errors	0	0	0
PCIe Unsupported Reque...	0	0	0
PCIe Bad DLLP	0	0	0
PCIe Bad TLP	0	0	0
PCIe NAK Received	0	0	0
PCIe NAK Sent	0	0	0
PCIe PEX Errors Recovery ...	171	147	171
PCIe Data Link Protocol E...	0	0	0

HWMonitor

File View Tools Help

Sensor	Value	Min	Max
PCIe Data Link Protocol E...	0	0	0
PCIe Surprise Down Errors	0	0	0
PCIe Complete Abort Err...	0	0	0
PCIe Poisoned TLP Receiv...	0	0	0
 Clocks			
Graphics	210.0 MHz	210.0 MHz	1500.0 MHz
Memory	405.0 MHz	405.0 MHz	6121.0 MHz
Video	555.0 MHz	555.0 MHz	1237.0 MHz
Crossbar	600.0 MHz	600.0 MHz	1335.0 MHz
System Domain	525.0 MHz	525.0 MHz	1185.0 MHz
Unknown #3	540.0 MHz	540.0 MHz	540.0 MHz
 Utilization			
GPU	0.0 %	0.0 %	73.0 %
Memory	10.9 %	10.3 %	11.9 %
Frame Buffer	0.0 %	0.0 %	7.0 %
Video Engine	0.0 %	0.0 %	1.0 %
Bus Interface	0.0 %	0.0 %	65.0 %
3D	0.0 %	0.0 %	18.9 %
Copy	0.0 %	0.0 %	38.0 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	316.2 MB	291.6 MB	358.6 MB
Shared Memory	68.7 MB	63.2 MB	125.0 MB
 Performance			
Power Limit	0	0	0
Temperature Limit	0	0	0
Reliable Voltage Limit	0	0	0
Operational Voltage Limit	0	0	0
Utilization Limit	1	1	1
SLI Synchronisation Limit	0	0	0
 Speed			
PCIe Read Throughput	0.19 MB/s	0.00 MB/s	106.88 MB/s
PCIe Write Throughput	0.00 MB/s	0.00 MB/s	2117.63 MB/s
PCIe Link Speed	2.50 GT/s	2.50 GT/s	16.00 GT/s
 Intel(R) UHD Graphics			
 Voltages			
GPU	0.745 V	0.732 V	1.042 V
 Clocks			
Graphics	600.0 MHz	550.0 MHz	1450.0 MHz
 Utilization			
GPU	1.6 %	0.0 %	32.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	1.6 %	0.0 %	32.0 %
Copy	0.0 %	0.0 %	0.0 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	0.0 MB	0.0 MB	0.0 MB
Shared Memory	451.1 MB	397.4 MB	567.8 MB

HWMonitor

File View Tools Help

Sensor	Value	Min	Max
Power Limit	0	0	0
Temperature Limit	0	0	0
Reliable Voltage Limit	0	0	0
Operational Voltage Limit	0	0	0
Utilization Limit	1	1	1
SLI Synchronisation Limit	0	0	0
Speed			
PCIe Read Throughput	0.00 MB/s	0.00 MB/s	106.88 MB/s
PCIe Write Throughput	0.00 MB/s	0.00 MB/s	2117.63 MB/s
PCIe Link Speed	2.50 GT/s	2.50 GT/s	16.00 GT/s
Intel(R) UHD Graphics			
Voltages			
GPU	0.745 V	0.732 V	1.042 V
Clocks			
Graphics	600.0 MHz	550.0 MHz	1450.0 MHz
Utilization			
GPU	0.4 %	0.0 %	32.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	0.4 %	0.0 %	32.0 %
Copy	0.0 %	0.0 %	0.0 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	0.0 MB	0.0 MB	0.0 MB
Shared Memory	425.3 MB	397.4 MB	567.8 MB
A32-K55			
Voltages			
Current Voltage	16.762 V	16.760 V	16.781 V
Powers			
Charge Rate	0.00 W	0.00 W	0.00 W
Capacities			
Designed Capacity	90006 mWh	90006 mWh	90006 mWh
Full Charge Capacity	64043 mWh	64043 mWh	64043 mWh
Current Capacity	64043 mWh	64043 mWh	64043 mWh
Levels			
Wear Level	28.8 %	28.8 %	28.8 %
Charge Level	100.0 %	100.0 %	100.0 %
VMware Virtual Ethernet Adapter for VMnet1			
VMware Virtual Ethernet Adapter for VMnet8			
MediaTek Wi-Fi 6 MT7921 Wireless LAN Card			
Speed			
Download Speed	1.51 Mbps	0.00 Mbps	5.14 Mbps
Upload Speed	0.08 Mbps	0.00 Mbps	0.47 Mbps
Realtek RTL8188FTV Wireless LAN 802.11n USB 2.0 Network Adapter			
Speed			
Download Speed	0.00 Mbps	0.00 Mbps	15.48 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.34 Mbps

Task 4 : Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Measures electrical values such as voltage, resistance, and continuity in computer or smartphone hardware.	Detects a faulty power adapter or broken wire by checking whether the correct voltage or continuity is present.
POST Card	Checks the Power-On Self-Test (POST) process of a motherboard and displays diagnostic codes during boot.	Helps identify RAM, CPU, or motherboard initialization problems when a computer fails to boot or shows no display.
Software Diagnostics	Monitors hardware information such as temperature, voltage, storage health, CPU/GPU usage, and SMART data .	Can detect CPU/GPU overheating or a failing HDD/SSD through abnormal temperatures or storage-health warnings.